

Mariana Paco

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Data analyst with 3+ years turning large, complex datasets into clear, decision-ready insights across business and research settings. Strong in SQL, Power BI, Tableau, and Python, with a track record of automating reporting and building dashboards that accelerate stakeholder decisions.

EDUCATION

University of California, San Diego Sep 2021 - Jun 2025

BS, Data Science

- Coursework: Data Management, Systems for Scalable Analytics, Statistical Methods, Probabilistic Modeling & Machine Learning, Data Mining & Recommender Systems

PROFESSIONAL EXPERIENCE

Animesh Ray's Lab, Keck Graduate Institute Sep 2023 - Present

Research Intern - Data Science

- Built classification models on datasets of >3,000 records by fine-tuning transformer models in a cloud/GPU environment, improving prediction accuracy from 55.2% to 80% and turning raw sequence data into decision-ready outputs.
- Reduced molecular dynamics simulation runtime by ~91% (from ~336 hours to <30 hours per protein) by optimizing GPU-enabled HPC workflows, enabling rapid evaluation of protein candidates and shortening project timelines.
- Standardized data-transfer and version-control (Git/GitHub) workflows across 2 projects, reducing data-handoff errors and making analyses fully reproducible.

Loam Bio Sep 2022 - Sep 2023

Bioinformatics Officer

- Reduced dataset retrieval time by 20% by re-architecting AWS S3 data-management protocols, enabling faster turnaround on multiple recurring analysis requests.
- Automated data-processing workflows in Python, reducing analysis turnaround from 5 days to 1 day and eliminating ~30 hours of repetitive manual work per cycle.
- Guided project decisions by building interactive Plotly/Dash dashboards for senior stakeholders, reducing ad-hoc reporting requests by 30%.

Keck Graduate Institute Jul 2022 - Sep 2022

Summer Research Assistant

- Enabled trend identification by developing Random Forest models (scikit-learn) on 1,000-record datasets, achieving 70% accuracy and informing next steps for modeling complex molecular interactions.
- Quantified intermolecular energy patterns across multiple datasets using Pandas for statistical analysis, delivering insights that supported research objectives.

RESEARCH & PROJECTS

ERP Business Intelligence Dashboard & Reporting Automation

- Integrated the Defontana ERP with interactive Plotly dashboards by automating API-based data ingestion and transformation pipelines, delivering client-specific payment reconciliation and order analytics that reduced reporting turnaround from 2 business days to a few hours.
- Developed tailored operational dashboards for finance and order-management teams, improving client responsiveness and accelerating data-driven decision-making.
- Queried relational databases with SQL to fulfill ad hoc reporting requests and support supervisory financial reviews.

ASPred: Identifying Antigen-Specific B-cell Receptors from V(D)J Sequences using LLMs

- Developed ASPred, a fine-tuned LLM predicting antigen-specific B-cell receptors from full repertoires without antigen-binding preselection, providing a scalable method for identifying stable, functional antibodies.

Ab-Affinity: LLM-Guided Affinity Maturation of Antibodies via Combinatorial Optimization

- Contributed to "Ab-Affinity," an LLM-based system designing synthetic antibodies with up to 160-fold improved binding affinity to the SARS-CoV-2 spike protein; manuscript in bioRxiv.

SKILLS & LANGUAGES

- **Languages & Analysis:** Python, R, SQL, Java, JavaScript, Bash Scripting
- **Data & BI:** Pandas, NumPy, Matplotlib, Plotly Express, Dash, Power BI, Tableau, Data Visualization, Scikit-learn, Keras, PyTorch
- **Cloud, ETL & Platforms:** AWS, Azure, Google Cloud (GCP), BigQuery, ETL & Data Warehousing, Python/Bash Pipelines, Git, Docker, Excel (IF formulas, VLOOKUP, Pivot Table Analysis)
- **Specialized & Languages:** Large Language Models, Causality with causal-learn, LaTeX; Spanish (fluent)